

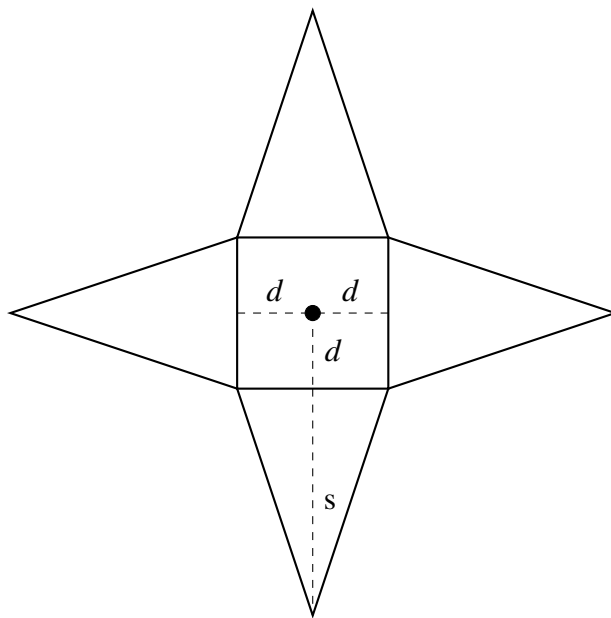
7th Grade Team Contest Solutions

IMSA *Mu Alpha Theta*

March 5, 2025

1. Since $75 = 3 \cdot 5^2$, to make this a perfect square multiply by $\boxed{3}$.
2. Multiplying out $11^7 = 19,487,171$. The digits $\boxed{4, 8, \text{ and } 9}$ are used only once.
3. Grant drove the first 10 miles in one hour, and the second 10 miles in $1/3$ hour, the average speed for the whole trip is $(20 \text{ miles}) / (4/3 \text{ hour}) = \boxed{15}$ miles per hour.
4. “Normal rules of operation” mean that multiple copies of the same operation are performed left-to-right. Starting on the left $1 \sqrt{1} = 1/2$, then $1/2 \star 1 = 1/(\frac{3}{2}) = 2/3$. The next operation yields $3/5$, then $5/8$, and so on. At each step, the old denominator is the new numerator, and the new denominator is the sum of the previous numerator and denominator. Continuing this pattern leads to a final answer of $\boxed{89/144}$.
5. The largest multiple of three that will fit into 13 is $12 = 4 \times 3$ and the largest that fits into 17 is $15 = 5 \times 3$. So $4 \times 5 = \boxed{20}$ servings can be cut from the large cake.
6. On any day, for each toy, she could either play with the toy or not that day. That’s two choices per toy. So however many combinations there would be with 8 toys, there are twice as many combinations with 9 toys. But similarly, there are twice as many combinations with 8 toys as there are with 7, and so on. So there are $2 \times 2 \times \cdots \times 2 = 2^9 = \boxed{512}$ combinations.
7. Basically, the question asks for the smallest n so that $\$1.50(1 + 2 + 3 + \cdots + n) \geq \100 . Trial and error shows that after 11 days, Alia has been paid a total of \$99. So she earns her hundredth dollar on day $\boxed{12}$.
8. A nifty fact about triangles inscribed in circles is that if one side of the triangle is a diameter of the circle, the opposite angle is a right angle. So $\angle C$ is a right angle, and by the Pythagorean theorem, $BD = \sqrt{15^2 + 20^2} = 25$. Then $\angle A$ is also a right angle, and the Pythagorean theorem applied to $\triangle ABD$ shows $AB = \sqrt{25^2 - 24^2} = \boxed{7}$.
9. Since i obeys the normal rules of arithmetic, let’s apply the distributive law (“FOIL”) to the first two terms: $(2+i)(3+4i) = 6 + 8i + 3i + 4i^2$. Now $i^2 = -1$ so this becomes $6 + 11i - 4 = 2 + 11i$. Likewise, the second two terms multiply together to become $(5+6i)(7+2i) = 35 + 10i + 42i - 12 = 33 + 52i$. Finally, multiplying these two gives $(2+11i)(33+52i) = 66 + 104i + 363i - 572 = \boxed{-506 + 467i}$.

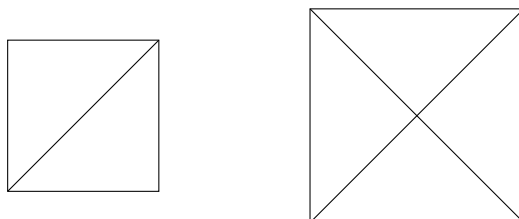
10. The first equation says that $a = 12$. The second then says $33 = 24 + 3b$, so $b = 3$. Finally, the third equation has $57 = 48 + 15 + 6c$, so $c = -1$. Adding, $a + b + c = 12 + 3 - 1 = \boxed{14}$.
11. You could take the square root of the first equation and do a lot of messy arithmetic, or try this clever approach. Square out the first equation to get $x^2 + 2xy + y^2 + 2x + 2y + 1 = 54$. Multiply out the second equation to get $xy + x + y + 1 = 15$ so $xy + x + y = 14$. Double this to get $2xy + 2x + 2y = 28$. Now subtract from the first equation to leave $x^2 + y^2 + 1 = 26$. Subtracting 1 from both sides of this equation reveals that $x^2 + y^2 = \boxed{25}$.
12. If Angela only used one bill, the most she could pay is \$100, which doesn't meet the conditions. Two bills that make at least \$150 are the two hundred dollar bills, but that is \$200, and the digits do not increase from left-to-right. Three bills that make at least \$150 could be \$201, \$205, or \$220, none of which have increasing digits. With four bills the total could be \$160, \$240, and a bunch of combinations between \$202 and \$226, none of which have increasing digits. Five bills won't do it either. But with six, Angela could pay $\boxed{\$235}$ which does fit all the criteria.
13. If there are n people at the table, they take $1 + 2 + \cdots + n = n(n + 1)/2$ pieces of candy. For this to be around 400, we need $n(n + 1) \approx 800$, so $n^2 \approx 800$. So n should be around 26 or 27. If there are 27 people at the table, the total amount of candy taken as the bag goes around the table is $27 \cdot 28/2 = 378$ pieces, meaning there are only 22 left for the first person, not quite enough to take the 28 pieces needed. So $\boxed{27}$ must be the number of people at the table.
14. Let the base of the pyramid have sides of length $2d$, and let s be the length of the segment from the midpoint of a side of the base up the face of the pyramid to the vertex at the top. Then imagine folding down the sides of the pyramid:



Now when the faces are folded back into the pyramid, the Pythagorean theorem shows that $s^2 = d^2 + 4d^2$, so $s = \sqrt{16 + d^2}$. The area of each triangular face of the pyramid is then

$d\sqrt{d^2 + 16}$ while the base has area $4d^2$. So the total surface area is $4d^2 + 4d\sqrt{d^2 + 16} = 16 + 16\sqrt{5}$. Either solving or guess-and-check reveals that $d = 2$. Then, since the volume of a pyramid is $\frac{1}{3}bh$ where b is the area of the base and h is the height, this evaluates to $\frac{1}{3} \cdot 4^2 \cdot 4 = \boxed{64/3}$.

15. Michael can combine two triangles into a square by placing their long sides together, or four by placing their short sides together:



Now these squares could be placed into 2×2 , 3×3 , etc. arrays to make larger squares. To make the most squares possible, Michael should use the smallest area squares he can until he runs out of triangles. So to make his squares will require 2, 4, 8, 16, 18, 32, 36, 50, 64, These are exactly the numbers that are either twice a square or four times a square. Since Michael only has 172 triangles, he must stop with the square that uses 50 triangles before he runs out of triangles. So he can make $\boxed{8}$ different sized squares.

16. Since we don't know which of $\oplus$ or $\otimes$ is performed first when they both appear, we will have to try all the possibilities and see which end up with integers.

Variation	Integer?
$(5 \otimes 10) \oplus 15 = \frac{50}{15} + 15 = 55/3$	No
$(10 \otimes 5) \oplus 15 = \frac{50}{15} + 15 = 55/3$	No
$(5 \otimes 15) \oplus 10 = \frac{75}{20} + 10 = 55/4$	No
$(15 \otimes 5) \oplus 10 = \frac{75}{20} + 10 = 55/4$	No
$(10 \otimes 15) \oplus 5 = \frac{150}{25} + 5 = 11$	Yes!
$(15 \otimes 10) \oplus 5 = \frac{150}{25} + 5 = 11$	Yes!
$5 \otimes (10 \oplus 15) = \frac{125}{30} = 25/6$	No
$5 \otimes (15 \oplus 10) = \frac{125}{50} = 25/6$	No
$10 \otimes (5 \oplus 15) = \frac{200}{30} = 20/3$	No
$10 \otimes (15 \oplus 5) = \frac{200}{30} = 20/3$	No
$15 \otimes (5 \oplus 15) = \frac{300}{35} = 60/7$	No
$15 \otimes (15 \oplus 5) = \frac{300}{35} = 60/7$	No

The only ones that are an integer have $a = 10$, $b = 15$, and $c = 5$, or a and b are reversed. Fortunately, the problem asks for their product, so order doesn't matter. The result is $10 \cdot 15 / 5 = \boxed{30}$.

17. Let the length of the side of the equilateral triangle be x , and let y be the short piece of the side of the square cut off by where the triangle intersects the side. Then from the Pythagorean theorem $2^2 + y^2 = x^2$ and $2(2 - y)^2 = x^2$. Thus, $8 - 8y + 2y^2 = 4 + y^2$ or $y^2 - 8y + 4 = 0$.

Using the quadratic formula, $y = \frac{8 \pm \sqrt{64 - 16}}{2} = 4 \pm \sqrt{12}$. The plus sign doesn't make sense, or y would be longer than the side of the square. So $y = 4 - 2\sqrt{3}$ and then $x^2 = 4 + (16 - 16\sqrt{3} + 12) = 32 - 16\sqrt{3}$. The area of an equilateral triangle whose sides have length x is $x^2 \sqrt{3}/4$, so multiplying the value just discovered for x^2 by $\sqrt{3}/4$ yields $8\sqrt{3} - 12$.

18. The problem asks to find x and y so that $77c(x) = c(y)$. Note that if x is a 1-digit number, then $c(x) = 11x$ while if x is a two-digit number then $c(x) = 101x$. Similarly for y , though y might also be a 3-digit number, in which case $c(y) = 1001y$.

Now y can't be 1-digit, because then $77c(x) = 11y$ or $7c(x) = y$ has no solutions since $c(x)$ has at least two digits. If y has two digits, then $77c(x) = 101y$. Because 101 is prime, neither 7 nor 11 factors into it, so y must be divisible by 77. The only two-digit number divisible by 77 is 77, so then $c(x) = 101$, but that is not a number written twice. So y can't be a two-digit number, either.

So y has three digits, and $77c(x) = 1001y$. Divide both sides of this equation by 77 to obtain $c(x) = 13y$. There are no 1-digit solutions for x , so x is a two-digit number, and therefore $101x = 13y$. Since 101 and 13 are both prime, this requires 13 to be a divisor of x and 101 to be a divisor of y . So x can be 13, 26, 39, 52, 78, or 91, and the sum of these is 364.

19. The clever trick for this problem is to rewrite the equation as $x^6 - y^2 = 127$, or $(x^3)^2 - y^2 = 127$. This is a difference of squares, which can be factored: $(x^3 - y)(x^3 + y) = 127$. Now 127 is prime, so $x^3 - y = 1$ and $x^3 + y = 127$. Adding these together gives $2x^3 = 128$ or $x^3 = 64$, so $x = 4$.

20. Since the angles of an equilateral triangle are 60° and the angles of a regular octagon are 135° , the triangle rotates 165° around its vertex each time it rolls to a new side of the octagon. This is $\frac{165}{360} = \frac{11}{24}$ of a complete revolution. Now a circle of radius 5 has a circumference of 10π , so $\frac{11}{24}$ of this is a distance of $\frac{55\pi}{12}$. This is how far the vertex P moves as the triangle makes its first roll.

But on the second roll, the triangle is rotating around vertex P , so it doesn't move at all! On the third roll, the same argument applies as on the first roll, and now the vertex P is not touching the octagon again. The next three rolls repeat this pattern. We would repeat the pattern again, but the triangle only needs the first two of those three rolls to return to its original position (although P is now touching the octagon). So altogether, the vertex P moved five times and was stationary three times. During those five rolls, P moved a total distance of $5 \cdot \frac{55\pi}{12} = 275\pi/12$.